Manage LDAP server configurations

ONTAP 9.14.1 REST API reference

NetApp
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Manage LDAP server configurations

Name-services LDAP endpoint overview

Overview

LDAP servers are used to centrally maintain user information. LDAP configurations must be set up to lookup information stored in the LDAP directory on the external LDAP servers. This API is used to retrieve and manage LDAP server configurations.

Retrieving LDAP information

The LDAP GET endpoint retrieves all of the LDAP configurations in the cluster.

Examples

Retrieving all of the fields for all LDAP configurations

```
# The API:
/api/name-services/ldap

# The call:
curl -X GET "https://<mgmt-ip>/api/name-services/ldap?fields=**" -H
"accept: application/hal+json"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "179d3c85-7053-11e8-b9b8-005056b41bd1",
      "name": "vs1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/179d3c85-7053-11e8-b9b8-005056b41bd1"
        }
      }
    },
    "servers": [
      "10.10.10.10",
      "domainB.example.com"
    ],
    "schema": "ad_idmu",
    "port": 389,
    "ldaps_enabled": false,
```

```
    "min_bind_level": "anonymous",
    "bind_dn": "cn=Administrators,cn=users,dc=domainA,dc=example,dc=com",
    "base_dn": "dc=domainA,dc=example,dc=com",
    "base_scope": "subtree",
    "use_start_tls": true,
    "session_security": "none",
    "referral_enabled": false,
    "bind_as_cifs_server": false,
    "query_timeout": 3,
    "is_owner": true,
    "user_scope": "subtree",
    "group_scope": "subtree",
    "netgroup_scope": "subtree",
    "is_netgroup_byhost_enabled": false,
    "netgroup_byhost_scope": "subtree",
    "group_membership_filter": "",
    "status": {
      "state": "down",
      "message": "The LDAP configuration is invalid. Verify that the AD
domain or servers are reachable and that the network configuration is
correct",
      "dn_message": [
        "No LDAP DN configured"
      ],
      "code": 4915258,
      "ipv4_state": "down",
      "ipv6_state": "down"
    },
    "try_channel_binding": true,
    "restrict_discovery_to_site" : false,
    "_links": {
      "self": {
        "href": "/api/name-services/ldap/179d3c85-7053-11e8-b9b8-
005056b41bd1"
      }
    }
  },
  {
    "svm": {
      "uuid": "6a52023b-7066-11e8-b9b8-005056b41bd1",
      "name": "vs2",
      "_links": {
        "self": {
          "href": "/api/svm/svms/6a52023b-7066-11e8-b9b8-005056b41bd1"
        }
      }
```

```
    },
    "ad_domain": "example.com",
    "schema": "rfc_2307",
    "port": 389,
    "ldaps_enabled": false,
    "min_bind_level": "simple",
    "bind_dn": "cn=Administrators,cn=users,dc=domainB,dc=example,dc=com",
    "base_dn": "dc=domainB,dc=example,dc=com",
    "base_scope": "subtree",
    "use_start_tls": true,
    "session_security": "sign",
    "referral_enabled": false,
    "bind_as_cifs_server": false,
    "query_timeout": 0,
    "is_owner": true,
    "user_scope": "subtree",
    "group_scope": "subtree",
    "netgroup_scope": "subtree",
    "is_netgroup_byhost_enabled": false,
    "netgroup_byhost_scope": "subtree",
    "group_membership_filter": "",
    "status": {
      "state": "up",
      "message": "Successfully connected to LDAP server \"172.20.192.44\".
Successfully connected to LDAP server
\"fd20:8b1e:b255:5056:999:d9:516c:bf69\".",
      "dn_message": [
        "All the configured DNs are available."
      ],
      "code": 0,
      "ipv4_state": "up",
      "ipv6_state": "up"
    },
    "try_channel_binding": true,
    "restrict_discovery_to_site": true,
    "_links": {
      "self": {
        "href": "/api/name-services/ldap/6a52023b-7066-11e8-b9b8-
005056b41bd1"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
```

```
      "href": "/api/name-services/ldap?fields=*"
    }
  }
  }
```

Retrieving all of the LDAP configurations that have the *use_start_tls* set to *true*

```
# The API:
/api/name-services/ldap

# The call:
curl -X GET "https://<mgmt-ip>/api/name-services/ldap?use_start_tls=true"
-H "accept: application/hal+json"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "6a52023b-7066-11e8-b9b8-005056b41bd1",
      "name": "vs2",
      "_links": {
        "self": {
          "href": "/api/svm/svms/6a52023b-7066-11e8-b9b8-005056b41bd1"
        }
      }
    },
    "use_start_tls": true,
    "_links": {
      "self": {
        "href": "/api/name-services/ldap/6a52023b-7066-11e8-b9b8-
005056b41bd1"
      }
    }
  }
],
"num_records": 1,
"_links": {
  "self": {
    "href": "/api/name-services/ldap?use_start_tls=true"
  }
}
}
```

Retrieving the LDAP configuration of a specific SVM

```
# The API:
/api/name-services/ldap/{svm.uuid}

# The call:
curl -X GET "https://<mgmt-ip>/api/name-services/ldap/179d3c85-7053-11e8-
b9b8-005056b41bd1" -H "accept: application/json"

# The response:
{
"svm": {
  "uuid": "179d3c85-7053-11e8-b9b8-005056b41bd1",
  "name": "vs1",
  "_links": {
    "self": {
      "href": "/api/svm/svms/179d3c85-7053-11e8-b9b8-005056b41bd1"
    }
  }
},
"servers": [
  "10.10.10.10",
  "domainB.example.com"
],
"schema": "ad_idmu",
"port": 389,
"ldaps_enabled": false,
"min_bind_level": "anonymous",
"bind_dn": "cn=Administrators,cn=users,dc=domainA,dc=example,dc=com",
"base_dn": "dc=domainA,dc=example,dc=com",
"base_scope": "subtree",
"use_start_tls": true,
"referral_enabled": false,
"session_security": "none",
"bind_as_cifs_server": true,
"query_timeout": 3,
"is_owner": true,
"try_channel_binding": true,
"restrict_discovery_to_site": false,
"_links": {
  "self": {
    "href": "/api/name-services/ldap/179d3c85-7053-11e8-b9b8-005056b41bd1"
  }
}
}
```

Retrieving all the fields of the LDAP configuration of a specific SVM

```
# The API:
/api/name-services/ldap/{svm.uuid}

# The call:
curl -X GET "https://<mgmt-ip>/api/name-services/ldap/179d3c85-7053-11e8-
b9b8-005056b41bd1?fields=**" -H "accept: application/json"

# The response:
{
"svm": {
  "uuid": "179d3c85-7053-11e8-b9b8-005056b41bd1",
  "name": "vs1",
  "_links": {
    "self": {
      "href": "/api/svm/svms/179d3c85-7053-11e8-b9b8-005056b41bd1"
    }
  }
},
"servers": [
  "10.10.10.10",
  "domainB.example.com"
],
"schema": "ad_idmu",
"port": 389,
"ldaps_enabled": false,
"min_bind_level": "anonymous",
"bind_dn": "cn=Administrators,cn=users,dc=domainA,dc=example,dc=com",
"base_dn": "dc=domainA,dc=example,dc=com",
"base_scope": "subtree",
"use_start_tls": true,
"referral_enabled": false,
"session_security": "none",
"bind_as_cifs_server": true,
"query_timeout": 3,
"is_owner": true,
"user_scope": "subtree",
"group_scope": "subtree",
"netgroup_scope": "subtree",
"is_netgroup_byhost_enabled": false,
"netgroup_byhost_scope": "subtree",
"group_membership_filter": "",
"try_channel_binding": true,
```

```
"restrict_discovery_to_site": false,
"status": {
  "state": "down",
  "message": "The LDAP configuration is invalid. Verify that the AD domain
or servers are reachable and that the network configuration is correct",
  "dn_message": [
    "No LDAP DN configured"
  ],
  "code": 4915258,
  "ipv4_state": "down",
  "ipv6_state": "down"
},
"_links": {
  "self": {
    "href": "/api/name-services/ldap/179d3c85-7053-11e8-b9b8-005056b41bd1"
  }
}
}
```


Retrieving the LDAP server status of a specific SVM

```
# The API:
/api/name-services/ldap/{svm.uuid}

# The call:
curl -X GET "https://<mgmt-ip>/api/name-services/ldap/9e4a2e3b-f66f-11ea-
aec8-0050568e155c?fields=status" -H "accept: application/json"

# The response:
{
"svm": {
  "uuid": "9e4a2e3b-f66f-11ea-aec8-0050568e155c",
  "name": "vs2"
},
"status": {
  "state": "up",
  "message": "Successfully connected to LDAP server \"172.20.192.44\".
Successfully connected to LDAP server
\"fd20:8b1e:b255:5056:999:d9:516c:bf69\".",
  "code": 0,
  "ipv4_state": "up",
  "ipv6_state": "up"
}
}
```

Retrieving all of the LDAP configurations that have the *restrict_discovery_to_site* set to *true*

```
# The API:
/api/name-services/ldap

# The call:
curl -X GET "https://<mgmt-ip>/api/name-
services/ldap?restrict_discovery_to_site=true" -H "accept:
application/hal+json"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "6a52023b-7066-11e8-b9b8-005056b41bd1",
      "name": "vs2",
      "_links": {
        "self": {
          "href": "/api/svm/svms/6a52023b-7066-11e8-b9b8-005056b41bd1"
        }
      }
    },
    "restrict_discovery_to_site": true,
    "_links": {
      "self": {
        "href": "/api/name-services/ldap/6a52023b-7066-11e8-b9b8-
005056b41bd1"
      }
    }
  }
],
"num_records": 1,
"_links": {
  "self": {
    "href": "/api/name-services/ldap?restrict_discovery_to_site=true"
  }
}
}
```

Creating an LDAP configuration

The LDAP POST endpoint creates an LDAP configuration for the specified SVM.

Examples

Creating an LDAP configuration with all the fields specified

```
# The API:
/api/name-services/ldap

# The call:
curl -X POST "https://<mgmt-ip>/api/name-services/ldap" -H "accept:
application/hal+json" -H "Content-Type: application/json" -d "{ \"svm\": {
\"uuid\": \"179d3c85-7053-11e8-b9b8-005056b41bd1\" }, \"servers\": [
\"10.10.10.10\", \"domainB.example.com\" ], \"schema\": \"ad_idmu\",
\"port\": 389, \"ldaps_enabled\": false, \"min_bind_level\":
\"anonymous\", \"bind_dn\":
\"cn=Administrators,cn=users,dc=domainA,dc=example,dc=com\",
\"bind_password\": \"abc\", \"base_dn\": \"dc=domainA,dc=example,dc=com\",
\"base_scope\": \"subtree\", \"use_start_tls\": false,
\"session_security\": \"none\", \"referral_enabled\": false,
\"bind_as_cifs_server\": false, \"query_timeout\": 4, \"user_dn\":
\"cn=abc,users,dc=com\", \"user_scope\": \"subtree\", \"group_dn\":
\"cn=abc,users,dc=com\", \"group_scope\": \"subtree\", \"netgroup_dn\":
\"cn=abc,users,dc=com\", \"netgroup_scope\": \"subtree\",
\"netgroup_byhost_dn\": \"cn=abc,users,dc=com\",
\"netgroup_byhost_scope\": \"subtree\", \"is_netgroup_byhost_enabled\":
false, \"group_membership_filter\": \"\", \"skip_config_validation\":
false } "
```

Creating an LDAP configuration with Active Directory domain and preferred Active Directory servers specified

```
# The API:
/api/name-services/ldap

# The call:
curl -X POST "https://<mgmt-ip>/api/name-services/ldap" -H "accept:
application/hal+json" -H "Content-Type: application/json" -d "{ \"svm\": {
\"name\": \"vs2\" }, \"ad_domain\": \"domainA.example.com\",
\"preferred_ad_servers\": [ \"11.11.11.11\" ], \"port\": 389,
\"ldaps_enabled\": false, \"bind_dn\":
\"cn=Administrators,cn=users,dc=domainA,dc=example,dc=com\",
\"bind_password\": \"abc\", \"base_dn\": \"dc=domainA,dc=example,dc=com\",
\"session_security\": \"none\", \"referral_enabled\": false,
\"query_timeout\": 3}"
```

Creating an LDAP configuration with a number of optional fields not specified

```
# The API:
/api/name-services/ldap

# The call:
curl -X POST "https://<mgmt-ip>/api/name-services/ldap" -H "accept:
application/hal+json" -H "Content-Type: application/json" -d "{ \"svm\": {
\"name\": \"vs2\" }, \"servers\": [ \"11.11.11.11\" ], \"port\": 389,
\"bind_dn\": \"cn=Administrators,cn=users,dc=domainA,dc=example,dc=com\",
\"bind_password\": \"abc\", \"base_dn\": \"dc=domainA,dc=example,dc=com\",
\"session_security\": \"none\"}"
```

Creating an LDAP configuration with Active Directory domain specified and *restrict_discovery_to_site* set to *true*

```
# The API:
/api/name-services/ldap

# The call:
curl -X POST "https://<mgmt-ip>/api/name-services/ldap" -H "accept:
application/hal+json" -H "Content-Type: application/json" -d "{ \"svm\": {
\"name\": \"vs2\" }, \"ad_domain\": \"example.com\", \"port\": 389,
\"bind_dn\": \"cn=Administrators,cn=users,dc=domainA,dc=example,dc=com\",
\"bind_password\": \"abc\", \"base_dn\": \"dc=domainA,dc=example,dc=com\",
\"session_security\": \"none\", \"restrict_discovery_to_site\": true}"
```

Updating an LDAP configuration

The LDAP PATCH endpoint updates the LDAP configuration for the specified SVM. The following example shows a PATCH operation:

```
# The API:
/api/name-services/ldap/{svm.uuid}

# The call:
curl -X PATCH "https://<mgmt-ip>/api/name-services/ldap/179d3c85-7053-
11e8-b9b8-005056b41bd1" -H "accept: application/json" -H "Content-Type:
application/json" -d "{ \"servers\": [ \"55.55.55.55\" ], \"schema\":
\"ad_idmu\", \"port\": 636, \"ldaps_enabled\": true, \"use_start_tls\":
false, \"referral_enabled\": false ,\"restrict_discovery_to_site\":
false}"
```

Deleting an LDAP configuration

The LDAP DELETE endpoint deletes the LDAP configuration for the specified SVM. The following example shows a DELETE operation:

```
# The API:
/api/name-services/ldap/{svm.uuid}

# The call:
curl -X DELETE "https://<mgmt-ip>/api/name-services/ldap/179d3c85-7053-
11e8-b9b8-005056b41bd1" -H "accept: application/hal+json"
```

Retrieve the LDAP configuration for all SVMs

GET `/name-services/ldap`

Introduced In: 9.6

Retrieves the LDAP configurations for all SVMs.

Related ONTAP commands

- `ldap show`
- `ldap check -vserver vs0`
- `ldap check-ipv6 -vserver vs0`

Important notes

- The status.code, status.dn_message, status.message, and status.state fields have the same status fields that are returned using the "ldap check" CLI command.
- Refer to the ipv4 or ipv6 objects available in the status field to get specific information about the code, dn_messages, or message and state information for ipv4 or ipv6.

Parameters

Name	Type	In	Required	Description
status.ipv6.state	string	query	False	Filter by status.ipv6.state • Introduced in: 9.14
status.ipv6.code	integer	query	False	Filter by status.ipv6.code • Introduced in: 9.14
status.ipv6.dn_messages	string	query	False	Filter by status.ipv6.dn_messages • Introduced in: 9.14
status.ipv6.message	string	query	False	Filter by status.ipv6.message • Introduced in: 9.14

Name	Type	In	Required	Description
status.state	string	query	False	Filter by status.state • Introduced in: 9.9
status.message	string	query	False	Filter by status.message • Introduced in: 9.9
status.dn_message	string	query	False	Filter by status.dn_message • Introduced in: 9.9
status.ipv4.state	string	query	False	Filter by status.ipv4.state • Introduced in: 9.14
status.ipv4.code	integer	query	False	Filter by status.ipv4.code • Introduced in: 9.14
status.ipv4.dn_messages	string	query	False	Filter by status.ipv4.dn_messages • Introduced in: 9.14
status.ipv4.message	string	query	False	Filter by status.ipv4.message • Introduced in: 9.14
status.code	integer	query	False	Filter by status.code • Introduced in: 9.9

Name	Type	In	Required	Description
status.ipv4_state	string	query	False	Filter by status.ipv4_state • Introduced in: 9.13
status.ipv6_state	string	query	False	Filter by status.ipv6_state • Introduced in: 9.13
netgroup_byhost_scope	string	query	False	Filter by netgroup_byhost_scope • Introduced in: 9.9
user_dn	string	query	False	Filter by user_dn • Introduced in: 9.9
netgroup_byhost_dn	string	query	False	Filter by netgroup_byhost_dn • Introduced in: 9.9
svm.uuid	string	query	False	Filter by svm.uuid • Introduced in: 9.7
svm.name	string	query	False	Filter by svm.name • Introduced in: 9.7
try_channel_binding	boolean	query	False	Filter by try_channel_binding • Introduced in: 9.10

Name	Type	In	Required	Description
session_security	string	query	False	Filter by session_security • Introduced in: 9.7
query_timeout	integer	query	False	Filter by query_timeout • Introduced in: 9.9
is_owner	boolean	query	False	Filter by is_owner • Introduced in: 9.9
group_membership_filter	string	query	False	Filter by group_membership_filter • Introduced in: 9.9
bind_as_cifs_server	boolean	query	False	Filter by bind_as_cifs_server • Introduced in: 9.9
user_scope	string	query	False	Filter by user_scope • Introduced in: 9.9
group_scope	string	query	False	Filter by group_scope • Introduced in: 9.9
bind_dn	string	query	False	Filter by bind_dn • Introduced in: 9.7

Name	Type	In	Required	Description
is_netgroup_byhost_enabled	boolean	query	False	Filter by is_netgroup_byhost_enabled • Introduced in: 9.9
port	integer	query	False	Filter by port • Introduced in: 9.7 • Max value: 65535 • Min value: 1
referral_enabled	boolean	query	False	Filter by referral_enabled • Introduced in: 9.9
restrict_discovery_to_site	boolean	query	False	Filter by restrict_discovery_to_site • Introduced in: 9.13
base_dn	string	query	False	Filter by base_dn • Introduced in: 9.7
group_dn	string	query	False	Filter by group_dn • Introduced in: 9.9
netgroup_scope	string	query	False	Filter by netgroup_scope • Introduced in: 9.9

Name	Type	In	Required	Description
netgroup_dn	string	query	False	Filter by netgroup_dn • Introduced in: 9.9
schema	string	query	False	Filter by schema • Introduced in: 9.7
min_bind_level	string	query	False	Filter by min_bind_level • Introduced in: 9.7
use_start_tls	boolean	query	False	Filter by use_start_tls • Introduced in: 9.7
base_scope	string	query	False	Filter by base_scope • Introduced in: 9.7
ad_domain	string	query	False	Filter by ad_domain • Introduced in: 9.7
preferred_ad_servers	string	query	False	Filter by preferred_ad_servers • Introduced in: 9.7
ldaps_enabled	boolean	query	False	Filter by ldaps_enabled • Introduced in: 9.9

Name	Type	In	Required	Description
servers	string	query	False	Filter by servers • Introduced in: 9.7
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Max value: 120 • Min value: 0 • Default value: 1
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
num_records	integer	Number of LDAP records.
records	array[ldap_service]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "ad_domain": "example.com",
    "base_dn": "dc=domainB,dc=example,dc=com",
    "base_scope": "base",
    "bind_dn":
"cn=Administrators,cn=users,dc=domainB,dc=example,dc=com",
    "bind_password": "abc",
    "group_dn": "cn=abc,users,dc=com",
    "group_membership_filter": "",
    "group_scope": "base",
    "min_bind_level": "anonymous",
    "netgroup_byhost_dn": "cn=abc,users,dc=com",
    "netgroup_byhost_scope": "base",
    "netgroup_dn": "cn=abc,users,dc=com",
    "netgroup_scope": "base",
    "port": 389,
    "preferred_ad_servers": {
    },
    "schema": "ad_idmu",
    "servers": {
    },
    "session_security": "none",
    "status": {
      "code": 65537300,
      "dn_message": {
      },
      "ipv4": {
        "code": 65537300,
        "dn_messages": {
```

```
        },
        "state": "up"
      },
      "ipv4_state": "up",
      "ipv6": {
        "code": 65537300,
        "dn_messages": {
        },
        "state": "up"
      },
      "ipv6_state": "up",
      "state": "up"
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "user_dn": "cn=abc,users,dc=com",
    "user_scope": "base"
  }
}
```


Error

```
Status: Default
```

Following error codes can be thrown as part of LDAP status information, if LDAP status is needed to be retrieved.

Error Code	Description
4915229	DNS resolution failed due to an internal error. Contact technical support if this issue persists
4915231	DNS resolution failed for one or more of the specified LDAP servers. Verify that a valid DNS server is configured
23724132	DNS resolution failed for all the specified LDAP servers. Verify that a valid DNS server is configured

Error Code	Description
4915258	The LDAP configuration is invalid. Verify that the Active Directory domain or servers are reachable and that the network configuration is correct
4915263	Failed to check the current status of LDAP server. Reason:<Reason for="" the="" failure=""></Reason>
4915234	The specified LDAP server or preferred Active Directory server is not supported because it is one of the following: multicast, loopback, 0.0.0.0, or broadcast
4915265	The specified bind password or bind DN is invalid
4915264	Certificate verification failed. Verify that a valid certificate is installed

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

_links

Name	Type	Description
self	href	

ipv4

Name	Type	Description
code	integer	Code corresponding to the error message. If there is no error, it is 0 to indicate success.
dn_messages	array[string]	
message	string	Provides additional details on the error.
state	string	Status of the LDAP service.

ipv6

Name	Type	Description
code	integer	Code corresponding to the error message. If there is no error, it is 0 to indicate success.
dn_messages	array[string]	
message	string	Provides additional details on the error.
state	string	Status of the LDAP service.

status

Name	Type	Description
code	integer	This field is no longer supported. Use ipv4.code or ipv6.code instead.
dn_message	array[string]	
ipv4	ipv4	
ipv4_state	string	This field is no longer supported. Use ipv4.state instead.
ipv6	ipv6	
ipv6_state	string	This field is no longer supported. Use ipv6.state instead.
message	string	This field is no longer supported. Use ipv4.message or ipv6.message instead.
state	string	The status of the LDAP service for the SVM. The LDAP service is up if either `ipv4_state` or `ipv6_state` is up. The LDAP service is down if both `ipv4_state` and `ipv6_state` are down.

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM. This field cannot be specified in a PATCH method.
uuid	string	The unique identifier of the SVM. This field cannot be specified in a PATCH method.

ldap_service

Name	Type	Description
_links	_links	
ad_domain	string	This parameter specifies the name of the Active Directory domain used to discover LDAP servers for use by this client. This is mutually exclusive with `servers` during POST and PATCH.
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_as_cifs_server	boolean	Specifies whether or not CIFS server's credentials are used to bind to the LDAP server.
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.
group_dn	string	Specifies the group Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for group lookups.
group_membership_filter	string	Specifies the custom filter used for group membership lookups from an LDAP server.

Name	Type	Description
group_scope	string	Specifies the default search scope for LDAP for group lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
is_netgroup_byhost_enabled	boolean	Specifies whether or not netgroup by host querying is enabled.
is_owner	boolean	Specifies whether or not the SVM owns the LDAP client configuration.
ldaps_enabled	boolean	Specifies whether or not LDAPS is enabled.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind
netgroup_byhost_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup by host lookups.

Name	Type	Description
netgroup_byhost_scope	string	Specifies the default search scope for LDAP for netgroup by host lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
netgroup_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup lookups.
netgroup_scope	string	Specifies the default search scope for LDAP for netgroup lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
port	integer	The port used to connect to the LDAP Servers.
preferred_ad_servers	array[string]	
query_timeout	integer	Specifies the maximum time to wait for a query response from the LDAP server, in seconds.
referral_enabled	boolean	Specifies whether or not LDAP referral is enabled.
restrict_discovery_to_site	boolean	Specifies whether or not LDAP server discovery is restricted to site-scope.

Name	Type	Description
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
skip_config_validation	boolean	Indicates whether or not the validation for the specified LDAP configuration is disabled.
status	status	
svm	svm	SVM, applies only to SVM-scoped objects.
try_channel_binding	boolean	Specifies whether or not channel binding is attempted in the case of TLS/LDAPS.
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.
user_dn	string	Specifies the user Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for user lookups.

Name	Type	Description
user_scope	string	Specifies the default search scope for LDAP for user lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Create the LDAP configuration for an SVM

POST `/name-services/ldap`

Introduced In: 9.6

Creates an LDAP configuration for an SVM.

Important notes

- Each SVM can have one LDAP configuration.
- The LDAP servers and Active Directory domain are mutually exclusive fields. These fields cannot be empty. At any point in time, either the LDAP servers or Active Directory domain must be populated.

- LDAP configuration with Active Directory domain cannot be created on an admin SVM.
- IPv6 must be enabled if IPv6 family addresses are specified.

The following parameters are optional:

- preferred AD servers
- schema
- port
- ldaps_enabled
- min_bind_level
- bind_password
- base_scope
- use_start_tls
- session_security
- referral_enabled
- bind_as_cifs_server
- query_timeout
- user_dn
- user_scope
- group_dn
- group_scope
- netgroup_dn
- netgroup_scope
- netgroup_byhost_dn
- netgroup_byhost_scope
- is_netgroup_byhost_enabled
- group_membership_filter
- skip_config_validation
- try_channel_binding
- restrict_discovery_to_site

Configuring more than one LDAP server is recommended to avoid a single point of failure. Both FQDNs and IP addresses are supported for the "servers" field. The Acitve Directory domain or LDAP servers are validated as part of this operation.

LDAP validation fails in the following scenarios:

1. The server does not have LDAP installed.
2. The server or Active Directory domain is invalid.
3. The server or Active Directory domain is unreachable.

Parameters

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Request Body

Name	Type	Description
_links	_links	
ad_domain	string	This parameter specifies the name of the Active Directory domain used to discover LDAP servers for use by this client. This is mutually exclusive with `servers` during POST and PATCH.
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_as_cifs_server	boolean	Specifies whether or not CIFS server's credentials are used to bind to the LDAP server.
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.

Name	Type	Description
group_dn	string	Specifies the group Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for group lookups.
group_membership_filter	string	Specifies the custom filter used for group membership lookups from an LDAP server.
group_scope	string	Specifies the default search scope for LDAP for group lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
is_netgroup_byhost_enabled	boolean	Specifies whether or not netgroup by host querying is enabled.
is_owner	boolean	Specifies whether or not the SVM owns the LDAP client configuration.
ldaps_enabled	boolean	Specifies whether or not LDAPS is enabled.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind
netgroup_byhost_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup by host lookups.

Name	Type	Description
netgroup_byhost_scope	string	Specifies the default search scope for LDAP for netgroup by host lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
netgroup_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup lookups.
netgroup_scope	string	Specifies the default search scope for LDAP for netgroup lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
port	integer	The port used to connect to the LDAP Servers.
preferred_ad_servers	array[string]	
query_timeout	integer	Specifies the maximum time to wait for a query response from the LDAP server, in seconds.
referral_enabled	boolean	Specifies whether or not LDAP referral is enabled.
restrict_discovery_to_site	boolean	Specifies whether or not LDAP server discovery is restricted to site-scope.

Name	Type	Description
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
skip_config_validation	boolean	Indicates whether or not the validation for the specified LDAP configuration is disabled.
status	status	
svm	svm	SVM, applies only to SVM-scoped objects.
try_channel_binding	boolean	Specifies whether or not channel binding is attempted in the case of TLS/LDAPS.
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.
user_dn	string	Specifies the user Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for user lookups.

Name	Type	Description
user_scope	string	Specifies the default search scope for LDAP for user lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN

Example request


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "ad_domain": "example.com",
  "base_dn": "dc=domainB,dc=example,dc=com",
  "base_scope": "base",
  "bind_dn": "cn=Administrators,cn=users,dc=domainB,dc=example,dc=com",
  "bind_password": "abc",
  "group_dn": "cn=abc,users,dc=com",
  "group_membership_filter": "",
  "group_scope": "base",
  "min_bind_level": "anonymous",
  "netgroup_byhost_dn": "cn=abc,users,dc=com",
  "netgroup_byhost_scope": "base",
  "netgroup_dn": "cn=abc,users,dc=com",
  "netgroup_scope": "base",
  "port": 389,
  "preferred_ad_servers": {
  },
  "schema": "ad_idmu",
  "servers": {
  },
  "session_security": "none",
  "status": {
    "code": 65537300,
    "dn_message": {
    },
    "ipv4": {
      "code": 65537300,
      "dn_messages": {
      },
      "state": "up"
    },
    "ipv4_state": "up",
    "ipv6": {
      "code": 65537300,
      "dn_messages": {
      },
      "state": "up"
    },
    "ipv6_state": "up",
```

```
    "state": "up"
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "user_dn": "cn=abc,users,dc=com",
  "user_scope": "base"
}
```


Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
num_records	integer	Number of LDAP records.
records	array[ldap_service]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "ad_domain": "example.com",
    "base_dn": "dc=domainB,dc=example,dc=com",
    "base_scope": "base",
    "bind_dn":
"cn=Administrators,cn=users,dc=domainB,dc=example,dc=com",
    "bind_password": "abc",
    "group_dn": "cn=abc,users,dc=com",
    "group_membership_filter": "",
    "group_scope": "base",
    "min_bind_level": "anonymous",
    "netgroup_byhost_dn": "cn=abc,users,dc=com",
    "netgroup_byhost_scope": "base",
    "netgroup_dn": "cn=abc,users,dc=com",
    "netgroup_scope": "base",
    "port": 389,
    "preferred_ad_servers": {
    },
    "schema": "ad_idmu",
    "servers": {
    },
    "session_security": "none",
    "status": {
      "code": 65537300,
      "dn_message": {
      },
      "ipv4": {
        "code": 65537300,
        "dn_messages": {
```

```
          },
          "state": "up"
        },
        "ipv4_state": "up",
        "ipv6": {
          "code": 65537300,
          "dn_messages": {
          },
          "state": "up"
        },
        "ipv6_state": "up",
        "state": "up"
      },
      "svm": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "svm1",
        "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
      },
      "user_dn": "cn=abc,users,dc=com",
      "user_scope": "base"
    }
  }
```


Headers

Name	Description	Type
Location	Useful for tracking the resource location	string

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
262186	LDAP Servers cannot be used with Active Directory domain and/or preferred Acti Directory servers
2621488	Invalid SVM context

Error Code	Description
2621706	The specified SVM UUID is incorrect for the specified SVM name
4915203	The specified LDAP schema does not exist
262222	The specified LDAP servers or preferred Active Directory servers contain duplicate server entries
4915229	DNS resolution failed due to an internal error. Contact technical support if this issue persists
4915231	DNS resolution failed for one or more of the specified LDAP servers. Verify that a valid DNS server is configured
23724132	DNS resolution failed for all the specified LDAP servers. Verify that a valid DNS server is configured
4915234	The specified LDAP server or preferred Active Directory server is not supported because it is one of the following: multicast, loopback, 0.0.0.0, or broadcast
4915248	LDAP servers cannot be empty or "-". Specified Active Directory domain is invalid because it is empty or "-" or it contains either the special characters or "-" at the start or end of the domain)
4915251	STARTTLS and LDAPS cannot be used together
4915257	The LDAP configuration is invalid. Verify that bind-dn and bind password are correct
4915258	The LDAP configuration is invalid. Verify that the Active Directory domain or servers are reachable and that the network configuration is correct
4915259	LDAP configurations with Active Directory domains are not supported on admin SVM.
4915265	The specified bind password or bind DN is invalid
4915264	Certificate verification failed. Verify that a valid certificate is installed
13434916	The SVM is in the process of being created. Wait a few minutes, and then try the command again.
23724130	Cannot use an IPv6 name server address because there are no IPv6 LIFs
4915252	LDAP Referral is not supported with STARTTLS, with session security levels sign, seal or with LDAPS.
4915266	LDAP site discovery restriction cannot be applied to a mixed version cluster.
656477	Need default site to be specified to enable site restriction.

Error Code	Description
4915206	CIFS server is not configured for the vserver. LDAP client configuration requires CIFS server for binding.
4915261	Cannot use port "389" when "ldaps_enabled" is "true".
4915255	Base DN specified in the LDAP client configuration is not available.

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

ipv4

Name	Type	Description
code	integer	Code corresponding to the error message. If there is no error, it is 0 to indicate success.
dn_messages	array[string]	
message	string	Provides additional details on the error.
state	string	Status of the LDAP service.

ipv6

Name	Type	Description
code	integer	Code corresponding to the error message. If there is no error, it is 0 to indicate success.
dn_messages	array[string]	
message	string	Provides additional details on the error.
state	string	Status of the LDAP service.

status

Name	Type	Description
code	integer	This field is no longer supported. Use ipv4.code or ipv6.code instead.

Name	Type	Description
dn_message	array[string]	
ipv4	ipv4	
ipv4_state	string	This field is no longer supported. Use ipv4.state instead.
ipv6	ipv6	
ipv6_state	string	This field is no longer supported. Use ipv6.state instead.
message	string	This field is no longer supported. Use ipv4.message or ipv6.message instead.
state	string	The status of the LDAP service for the SVM. The LDAP service is up if either `ipv4_state` or `ipv6_state` is up. The LDAP service is down if both `ipv4_state` and `ipv6_state` are down.

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM. This field cannot be specified in a PATCH method.
uuid	string	The unique identifier of the SVM. This field cannot be specified in a PATCH method.

ldap_service

Name	Type	Description
_links	_links	

Name	Type	Description
ad_domain	string	This parameter specifies the name of the Active Directory domain used to discover LDAP servers for use by this client. This is mutually exclusive with `servers` during POST and PATCH.
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_as_cifs_server	boolean	Specifies whether or not CIFS server's credentials are used to bind to the LDAP server.
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.
group_dn	string	Specifies the group Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for group lookups.
group_membership_filter	string	Specifies the custom filter used for group membership lookups from an LDAP server.

Name	Type	Description
group_scope	string	Specifies the default search scope for LDAP for group lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
is_netgroup_byhost_enabled	boolean	Specifies whether or not netgroup by host querying is enabled.
is_owner	boolean	Specifies whether or not the SVM owns the LDAP client configuration.
ldaps_enabled	boolean	Specifies whether or not LDAPS is enabled.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind
netgroup_byhost_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup by host lookups.

Name	Type	Description
netgroup_byhost_scope	string	Specifies the default search scope for LDAP for netgroup by host lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
netgroup_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup lookups.
netgroup_scope	string	Specifies the default search scope for LDAP for netgroup lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
port	integer	The port used to connect to the LDAP Servers.
preferred_ad_servers	array[string]	
query_timeout	integer	Specifies the maximum time to wait for a query response from the LDAP server, in seconds.
referral_enabled	boolean	Specifies whether or not LDAP referral is enabled.
restrict_discovery_to_site	boolean	Specifies whether or not LDAP server discovery is restricted to site-scope.

Name	Type	Description
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
skip_config_validation	boolean	Indicates whether or not the validation for the specified LDAP configuration is disabled.
status	status	
svm	svm	SVM, applies only to SVM-scoped objects.
try_channel_binding	boolean	Specifies whether or not channel binding is attempted in the case of TLS/LDAPS.
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.
user_dn	string	Specifies the user Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for user lookups.

Name	Type	Description
user_scope	string	Specifies the default search scope for LDAP for user lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Delete the LDAP configuration for an SVM

DELETE `/name-services/ldap/{svm.uuid}`

Introduced In: 9.6

Deletes the LDAP configuration of the specified SVM. LDAP can be removed as a source from the ns-switch if LDAP is not used as a source for lookups.

Parameters

Name	Type	In	Required	Description
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.

Response

```
Status: 200, Ok
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve the LDAP configuration for an SVM

GET `/name-services/ldap/{svm.uuid}`

Introduced In: 9.6

Retrieves LDAP configuration for an SVM. All parameters for the LDAP configuration are displayed by default.

Related ONTAP commands

- `ldap show`
- `ldap check -vserver vs0`
- `ldap check-ipv6 -vserver vs0`

Important notes

- The status.code, status.dn_message, status.message, and status.state fields have the same status fields that are returned using the "ldap check" CLI command.
- Refer to the ipv4 or ipv6 objects available in the status field to get specific information about the code, dn_messages, or message and state information for ipv4 or ipv6.

Parameters

Name	Type	In	Required	Description
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
ad_domain	string	This parameter specifies the name of the Active Directory domain used to discover LDAP servers for use by this client. This is mutually exclusive with `servers` during POST and PATCH.
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_as_cifs_server	boolean	Specifies whether or not CIFS server's credentials are used to bind to the LDAP server.
bind_dn	string	Specifies the user that binds to the LDAP servers.

Name	Type	Description
bind_password	string	Specifies the bind password for the LDAP servers.
group_dn	string	Specifies the group Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for group lookups.
group_membership_filter	string	Specifies the custom filter used for group membership lookups from an LDAP server.
group_scope	string	Specifies the default search scope for LDAP for group lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
is_netgroup_byhost_enabled	boolean	Specifies whether or not netgroup by host querying is enabled.
is_owner	boolean	Specifies whether or not the SVM owns the LDAP client configuration.
ldaps_enabled	boolean	Specifies whether or not LDAPS is enabled.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind
netgroup_byhost_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup by host lookups.

Name	Type	Description
netgroup_byhost_scope	string	Specifies the default search scope for LDAP for netgroup by host lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
netgroup_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup lookups.
netgroup_scope	string	Specifies the default search scope for LDAP for netgroup lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
port	integer	The port used to connect to the LDAP Servers.
preferred_ad_servers	array[string]	
query_timeout	integer	Specifies the maximum time to wait for a query response from the LDAP server, in seconds.
referral_enabled	boolean	Specifies whether or not LDAP referral is enabled.
restrict_discovery_to_site	boolean	Specifies whether or not LDAP server discovery is restricted to site-scope.

Name	Type	Description
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
skip_config_validation	boolean	Indicates whether or not the validation for the specified LDAP configuration is disabled.
status	status	
svm	svm	SVM, applies only to SVM-scoped objects.
try_channel_binding	boolean	Specifies whether or not channel binding is attempted in the case of TLS/LDAPS.
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.
user_dn	string	Specifies the user Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for user lookups.

Name	Type	Description
user_scope	string	Specifies the default search scope for LDAP for user lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN

Example response


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "ad_domain": "example.com",
  "base_dn": "dc=domainB,dc=example,dc=com",
  "base_scope": "base",
  "bind_dn": "cn=Administrators,cn=users,dc=domainB,dc=example,dc=com",
  "bind_password": "abc",
  "group_dn": "cn=abc,users,dc=com",
  "group_membership_filter": "",
  "group_scope": "base",
  "min_bind_level": "anonymous",
  "netgroup_byhost_dn": "cn=abc,users,dc=com",
  "netgroup_byhost_scope": "base",
  "netgroup_dn": "cn=abc,users,dc=com",
  "netgroup_scope": "base",
  "port": 389,
  "preferred_ad_servers": {
  },
  "schema": "ad_idmu",
  "servers": {
  },
  "session_security": "none",
  "status": {
    "code": 65537300,
    "dn_message": {
    },
    "ipv4": {
      "code": 65537300,
      "dn_messages": {
      },
      "state": "up"
    },
    "ipv4_state": "up",
    "ipv6": {
      "code": 65537300,
      "dn_messages": {
      },
      "state": "up"
    },
    "ipv6_state": "up",
```

```
    "state": "up"
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "user_dn": "cn=abc,users,dc=com",
  "user_scope": "base"
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	returned_error	

Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```


Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

ipv4

Name	Type	Description
code	integer	Code corresponding to the error message. If there is no error, it is 0 to indicate success.
dn_messages	array[string]	
message	string	Provides additional details on the error.
state	string	Status of the LDAP service.

ipv6

Name	Type	Description
code	integer	Code corresponding to the error message. If there is no error, it is 0 to indicate success.
dn_messages	array[string]	
message	string	Provides additional details on the error.
state	string	Status of the LDAP service.

status

Name	Type	Description
code	integer	This field is no longer supported. Use ipv4.code or ipv6.code instead.

Name	Type	Description
dn_message	array[string]	
ipv4	ipv4	
ipv4_state	string	This field is no longer supported. Use ipv4.state instead.
ipv6	ipv6	
ipv6_state	string	This field is no longer supported. Use ipv6.state instead.
message	string	This field is no longer supported. Use ipv4.message or ipv6.message instead.
state	string	The status of the LDAP service for the SVM. The LDAP service is up if either `ipv4_state` or `ipv6_state` is up. The LDAP service is down if both `ipv4_state` and `ipv6_state` are down.

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM. This field cannot be specified in a PATCH method.
uuid	string	The unique identifier of the SVM. This field cannot be specified in a PATCH method.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update the LDAP configuration for an SVM

PATCH `/name-services/ldap/{svm.uuid}`

Introduced In: 9.6

Updates an LDAP configuration of an SVM.

Important notes

- Both mandatory and optional parameters of the LDAP configuration can be updated.
- The LDAP servers and Active Directory domain are mutually exclusive fields. These fields cannot be empty. At any point in time, either the LDAP servers or Active Directory domain must be populated.
- IPv6 must be enabled if IPv6 family addresses are specified.

Configuring more than one LDAP server is recommended to avoid a sinlge point of failure. Both FQDNs and IP addresses are supported for the "servers" field. The Active Directory domain or LDAP servers are validated as part of this operation.

LDAP validation fails in the following scenarios:

1. The server does not have LDAP installed.
2. The server or Active Directory domain is invalid.
3. The server or Active Directory domain is unreachable

Parameters

Name	Type	In	Required	Description
svm.uuid	string	path	True	UUID of the SVM to which this object belongs.

Request Body

Name	Type	Description
_links	_links	
ad_domain	string	This parameter specifies the name of the Active Directory domain used to discover LDAP servers for use by this client. This is mutually exclusive with `servers` during POST and PATCH.
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_as_cifs_server	boolean	Specifies whether or not CIFS server's credentials are used to bind to the LDAP server.
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.
group_dn	string	Specifies the group Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for group lookups.
group_membership_filter	string	Specifies the custom filter used for group membership lookups from an LDAP server.

Name	Type	Description
group_scope	string	Specifies the default search scope for LDAP for group lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
is_netgroup_byhost_enabled	boolean	Specifies whether or not netgroup by host querying is enabled.
is_owner	boolean	Specifies whether or not the SVM owns the LDAP client configuration.
ldaps_enabled	boolean	Specifies whether or not LDAPS is enabled.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind
netgroup_byhost_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup by host lookups.
netgroup_byhost_scope	string	Specifies the default search scope for LDAP for netgroup by host lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN

Name	Type	Description
netgroup_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup lookups.
netgroup_scope	string	Specifies the default search scope for LDAP for netgroup lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
port	integer	The port used to connect to the LDAP Servers.
preferred_ad_servers	array[string]	
query_timeout	integer	Specifies the maximum time to wait for a query response from the LDAP server, in seconds.
referral_enabled	boolean	Specifies whether or not LDAP referral is enabled.
restrict_discovery_to_site	boolean	Specifies whether or not LDAP server discovery is restricted to site-scope.
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	

Name	Type	Description
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
skip_config_validation	boolean	Indicates whether or not the validation for the specified LDAP configuration is disabled.
status	status	
svm	svm	SVM, applies only to SVM-scoped objects.
try_channel_binding	boolean	Specifies whether or not channel binding is attempted in the case of TLS/LDAPS.
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.
user_dn	string	Specifies the user Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for user lookups.
user_scope	string	Specifies the default search scope for LDAP for user lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN

Example request


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "ad_domain": "example.com",
  "base_dn": "dc=domainB,dc=example,dc=com",
  "base_scope": "base",
  "bind_dn": "cn=Administrators,cn=users,dc=domainB,dc=example,dc=com",
  "bind_password": "abc",
  "group_dn": "cn=abc,users,dc=com",
  "group_membership_filter": "",
  "group_scope": "base",
  "min_bind_level": "anonymous",
  "netgroup_byhost_dn": "cn=abc,users,dc=com",
  "netgroup_byhost_scope": "base",
  "netgroup_dn": "cn=abc,users,dc=com",
  "netgroup_scope": "base",
  "port": 389,
  "preferred_ad_servers": {
  },
  "schema": "ad_idmu",
  "servers": {
  },
  "session_security": "none",
  "status": {
    "code": 65537300,
    "dn_message": {
    },
    "ipv4": {
      "code": 65537300,
      "dn_messages": {
      },
      "state": "up"
    },
    "ipv4_state": "up",
    "ipv6": {
      "code": 65537300,
      "dn_messages": {
      },
      "state": "up"
    },
    "ipv6_state": "up",
```

```
    "state": "up"
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "user_dn": "cn=abc,users,dc=com",
  "user_scope": "base"
}
```


Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
262186	LDAP Servers cannot be used with Active Directory domain and/or preferred Active Directory servers
2621488	Invalid SVM context
2621706	The specified SVM UUID is incorrect for the specified SVM name
4915203	The specified LDAP schema does not exist
262222	The specified LDAP servers or preferred Active Directory servers contain duplicate server entries
4915229	DNS resolution failed due to an internal error. Contact technical support if this issue persists
4915231	DNS resolution failed for one or more of the specified LDAP servers. Verify that a valid DNS server is configured
23724132	DNS resolution failed for all the specified LDAP servers. Verify that a valid DNS server is configured

Error Code	Description
4915234	The specified LDAP server or preferred Active Directory server is not supported because it is one of the following: multicast, loopback, 0.0.0.0, or broadcast
4915248	LDAP servers cannot be empty or "-". Specified Active Directory domain is invalid because it is empty or "-" or it contains either the special characters or "-" at the start or end of the domain.
4915251	STARTTLS and LDAPS cannot be used together
4915257	The LDAP configuration is invalid. Verify that the distinguished names and bind password are correct
4915258	The LDAP configuration is invalid. Verify that the Active Directory domain or servers are reachable and that the network configuration is correct
4915259	LDAP configurations with Active Directory domains are not supported on admin SVM.
23724130	Cannot use an IPv6 name server address because there are no IPv6 LIFs
4915252	LDAP Referral is not supported with STARTTLS, with session security levels sign, seal or with LDAPS.
4915266	LDAP site discovery restriction cannot be applied to a mixed version cluster.
656477	Need default site to be specified to enable site restriction.
4915206	CIFS server is not configured for the vserver. LDAP client configuration requires CIFS server for binding.
4915244	RPC failure occurred during validation of the LDAP configuration.

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

ipv4

Name	Type	Description
code	integer	Code corresponding to the error message. If there is no error, it is 0 to indicate success.
dn_messages	array[string]	
message	string	Provides additional details on the error.
state	string	Status of the LDAP service.

ipv6

Name	Type	Description
code	integer	Code corresponding to the error message. If there is no error, it is 0 to indicate success.
dn_messages	array[string]	
message	string	Provides additional details on the error.
state	string	Status of the LDAP service.

status

Name	Type	Description
code	integer	This field is no longer supported. Use ipv4.code or ipv6.code instead.

Name	Type	Description
dn_message	array[string]	
ipv4	ipv4	
ipv4_state	string	This field is no longer supported. Use ipv4.state instead.
ipv6	ipv6	
ipv6_state	string	This field is no longer supported. Use ipv6.state instead.
message	string	This field is no longer supported. Use ipv4.message or ipv6.message instead.
state	string	The status of the LDAP service for the SVM. The LDAP service is up if either `ipv4_state` or `ipv6_state` is up. The LDAP service is down if both `ipv4_state` and `ipv6_state` are down.

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM. This field cannot be specified in a PATCH method.
uuid	string	The unique identifier of the SVM. This field cannot be specified in a PATCH method.

ldap_service

Name	Type	Description
_links	_links	

Name	Type	Description
ad_domain	string	This parameter specifies the name of the Active Directory domain used to discover LDAP servers for use by this client. This is mutually exclusive with `servers` during POST and PATCH.
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_as_cifs_server	boolean	Specifies whether or not CIFS server's credentials are used to bind to the LDAP server.
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.
group_dn	string	Specifies the group Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for group lookups.
group_membership_filter	string	Specifies the custom filter used for group membership lookups from an LDAP server.

Name	Type	Description
group_scope	string	Specifies the default search scope for LDAP for group lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
is_netgroup_byhost_enabled	boolean	Specifies whether or not netgroup by host querying is enabled.
is_owner	boolean	Specifies whether or not the SVM owns the LDAP client configuration.
ldaps_enabled	boolean	Specifies whether or not LDAPS is enabled.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind
netgroup_byhost_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup by host lookups.

Name	Type	Description
netgroup_byhost_scope	string	Specifies the default search scope for LDAP for netgroup by host lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
netgroup_dn	string	Specifies the netgroup Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for netgroup lookups.
netgroup_scope	string	Specifies the default search scope for LDAP for netgroup lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
port	integer	The port used to connect to the LDAP Servers.
preferred_ad_servers	array[string]	
query_timeout	integer	Specifies the maximum time to wait for a query response from the LDAP server, in seconds.
referral_enabled	boolean	Specifies whether or not LDAP referral is enabled.
restrict_discovery_to_site	boolean	Specifies whether or not LDAP server discovery is restricted to site-scope.

Name	Type	Description
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
skip_config_validation	boolean	Indicates whether or not the validation for the specified LDAP configuration is disabled.
status	status	
svm	svm	SVM, applies only to SVM-scoped objects.
try_channel_binding	boolean	Specifies whether or not channel binding is attempted in the case of TLS/LDAPS.
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.
user_dn	string	Specifies the user Distinguished Name (DN) that is used as the starting point in the LDAP directory tree for user lookups.

Name	Type	Description
user_scope	string	Specifies the default search scope for LDAP for user lookups: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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